Assignment 6 12209 0095 杜奕璇

Q1. $a_1^T a_2 = \alpha \|a_2\|$

$\|a_1\|^2 \|a_2\|^2 - (a_1^T a_2)^2 = (\|a_1\|^2 - \alpha^2)\, a_2^2 = h^2 \|a_2\|^2$

Q2. $u^T v = a^2 + b - b^2 = 0$

$a = b+2 \Rightarrow (b+2)^2 + b - b^2 = 0$

$\Rightarrow a = -\frac{1}{2} \quad b = -\frac{5}{2}$

Q3. (a) False Let $V = (x, 0, 0)$. $W = (0, y, 0)$.

V and W are subspaces of R^3

$V^\perp = (0, y, z) \quad W^\perp = (x, 0, z)$

$\alpha = (0, 0, 1) \in V^\perp$ and $\alpha \in W^\perp$, but $\alpha^T \alpha = 1 \neq 0$.

thus $V^\perp$ and $W^\perp$ are not orthogonal

(b) False Let $Z = V \neq 0$ and V, W are orthogonal, thus Z, W are orthogonal

but V and V are not orthogonal.

Q4 (a) $A^T = (xy^T)^T + (yx^T)^T = yx^T + xy^T = A$.

(b) $\forall \alpha \in S^\perp \quad x^T \alpha = y^T \alpha = 0 \quad A\alpha = yx^T\alpha + xy^T\alpha = 0$.

$\therefore \alpha \in NULL(A) \quad S^\perp \subseteq NULL(A)$

$\forall \alpha \in NULL(A) \quad A\alpha = yx^T\alpha + xy^T\alpha = (x^T\alpha)\, y + (y^T\alpha)\, x = 0$

$\because x, y$ are linearly independent

$\therefore x^T\alpha = 0, \quad y^T\alpha = 0$

$\therefore \alpha \in S^\perp \quad NULL(A) \subseteq S^\perp$

$\therefore NULL(A) = S^\perp$

Q5. (a) $\|X\|_2 = \sqrt{\sum_{i=1}^n |x_i|^2} \geq \sqrt{\max_{i=1,\cdots,n} |x_i|^2} = \max_{i=1,\cdots,n} |x_i| = \|X\|_\infty$

(b) $\|X\|_1 = \sum_{i=1}^n |x_i| > 0$

$\|X\|_2^2 = \sum_{i=1}^n |x_i|^2 \leq \sum_{i=1}^n \sum_{j=1}^n |x_i||x_j| = \left(\sum_{i=1}^n |x_i|\right)\left(\sum_{j=1}^n |x_j|\right) = \|X\|_1^2$

$\therefore \|X\|_2 \leq \|X\|_1$.

Q6. $\begin{cases} \frac{1}{2}a + \frac{1}{2}b + \frac{1}{2}c + \frac{1}{2}d = 0 \\ \frac{1}{2}a - \frac{1}{2}b + \frac{1}{2}c - \frac{1}{2}d = 0 \\ \frac{1}{\sqrt{2}}a - \frac{1}{\sqrt{2}}c = 0 \\ a^2 + b^2 + c^2 + d^2 = 1 \end{cases} \Rightarrow \begin{cases} a = 0 \\ b = \frac{1}{\sqrt{2}} \\ c = 0 \\ d = -\frac{1}{\sqrt{2}} \end{cases}$

Q7. $\cos\theta = \frac{v^T w}{\|v\| \cdot \|w\|} = \frac{84}{\sqrt{14} \times 6\sqrt{14}} = 1$

$\theta = 0$

Q8. $p = \frac{x^T y}{\|y\|^2} y = \frac{8}{2}(1,1)^T = (4,4)^T$

$x - p = (3,5) - (4,4) = (-1,1)$

$(x-p)^T p = -4 + 4 = 0$ $\therefore$ $x-p$, p are orthogonal

Q9. (a) $X^T X = X_1^2 + X_2^2 \geq 0$

(b) $X^T y = (x_1, x_2)\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = x_1 y_1 + x_2 y_2 = (y_1, y_2)\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = y^T x$

(c) $X^T(y+z) = (x_1, x_2)\begin{pmatrix} y_1+z_1 \\ y_2+z_2 \end{pmatrix} = x_1(y_1+z_1) + x_2(y_2+z_2)$

$= (x_1, x_2)\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} + (x_1, x_2)\begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = X^T y + X^T z$

Q10 (a) $\alpha = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ $\alpha \in NULL(A) \Leftrightarrow A\alpha = 0$

$\Leftrightarrow \begin{cases} x_1 a_1 + x_2 a_2 + x_3 a_3 = 0 \\ y_1 a_1 + y_2 a_2 + y_3 a_3 = 0 \end{cases}$

$\Leftrightarrow X^T \alpha = 0, \; y^T \alpha = 0$

$\Leftrightarrow \alpha \in S^\perp$

(b) $\alpha \in S^\perp \Rightarrow \alpha \in NULL(A)$.

$\begin{pmatrix} -1 & 2 & 1 \\ 1 & -1 & 0 \end{pmatrix}\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ $\alpha = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$ (not unique)

Q11 (a) $A^T A = \begin{pmatrix} 1 & 2 & -1 \\ 4 & 1 & -3 \end{pmatrix}\begin{pmatrix} 1 & 4 \\ 2 & 1 \\ -1 & -3 \end{pmatrix} = \begin{pmatrix} 6 & 9 \\ 9 & 26 \end{pmatrix}$ $A^T b = \begin{pmatrix} 1 & 2 & -1 \\ 4 & 1 & -3 \end{pmatrix}\begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 6 \\ 11 \end{pmatrix}$

$A^T A x = A^T b \Rightarrow \begin{pmatrix} 6 & 9 \\ 9 & 26 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 6 \\ 11 \end{pmatrix} \Rightarrow x = \begin{pmatrix} \frac{19}{25} \\ \frac{4}{25} \end{pmatrix}$

(b) $A^T A = \begin{pmatrix} 1 & -1 & 1 \\ 1 & 3 & 2 \\ 3 & 1 & 4 \end{pmatrix}\begin{pmatrix} 1 & 1 & 3 \\ -1 & 3 & 1 \\ 1 & 2 & 4 \end{pmatrix} = \begin{pmatrix} 3 & 0 & 6 \\ 0 & 14 & 14 \\ 6 & 14 & 26 \end{pmatrix}$ $A^T b = \begin{pmatrix} 1 & -1 & 1 \\ 1 & 3 & 2 \\ 3 & 1 & 4 \end{pmatrix}\begin{pmatrix} -2 \\ 0 \\ 4 \end{pmatrix} = \begin{pmatrix} 2 \\ 6 \\ 10 \end{pmatrix}$

$\begin{pmatrix} 3 & 0 & 6 \\ 0 & 14 & 14 \\ 6 & 14 & 26 \end{pmatrix} x = \begin{pmatrix} 2 \\ 6 \\ 10 \end{pmatrix}$ $x = k_1\begin{pmatrix} \frac{2}{3} \\ \frac{3}{7} \\ 0 \end{pmatrix} + k_2\begin{pmatrix} \frac{8}{3} \\ \frac{10}{7} \\ -1 \end{pmatrix}$ $(k_1, k_2 \in \mathbb{R})$

Q12 (a) $P^2 = A(A^T A)^{-1} A^T A (A^T A)^{-1} A^T = A(A^T A)^{-1} A^T = P$

(b) if $P^{k-1} = P$ $(k \geq 2)$. then $P^k = P P^{k-1} = P P = P$

since $P^2 = P$. we have $P^k = P$ $(k = 1, 2, \dots)$

(c) $P^T = (A(A^TA)^{-1}A^T)^T = A((A^TA)^{-1})^TA^T$

$= A((A^TA)^T)^{-1}A^T = A(A^TA)^{-1}A^T = P$

$\therefore P$ is symmetric

Q13 (a) $\cos\theta = \frac{x^Ty}{\|x\|\|y\|} = \frac{6}{\sqrt{6}\times\sqrt{21}} = \frac{2}{\sqrt{14}}$ $\theta = \cos^{-1}\frac{\sqrt{14}}{7}$

(b) $p = \frac{x^Ty}{\|y\|^2}y = \frac{6}{21}(4,1,2,0)^T = (\frac{8}{7}, \frac{2}{7}, \frac{4}{7}, 0)$

(c) $x-p = (-\frac{1}{7}, \frac{12}{7}, -\frac{4}{7}, 1)$

$(x-p)^Tp = -\frac{8}{49} + \frac{24}{49} - \frac{16}{49} = 0$

(d) $\|x-p\|_2 = \frac{\sqrt{210}}{7}$ $\|p\|_2 = \frac{2\sqrt{21}}{7}$ $\|x\| = \sqrt{6}$

$(\|x-p\|_2)^2 + (\|p\|_2)^2 = \frac{210+84}{49} = 6 = \|x\|^2$

Q14. (a) $u_1^Tu_2 = \frac{\sqrt{2}}{9} + \frac{\sqrt{2}}{9} - \frac{2\sqrt{2}}{9} = 0$

$u_1^Tu_3 = \frac{1}{6} - \frac{1}{6} = 0$

$u_2^Tu_3 = \frac{\sqrt{2}}{3} - \frac{\sqrt{2}}{3} = 0$

$\|u_1\| = \sqrt{u_1^Tu_1} = 1$ $\|u_2\| = 1$ $\|u_3\| = 1$

(b) $\langle x, u_1\rangle = -\frac{\sqrt{2}}{3}$ $\langle x, u_2\rangle = \frac{5}{3}$ $\langle x, u_3\rangle = 0$

$\therefore x = -\frac{\sqrt{2}}{3}u_1 + \frac{5}{3}u_2$

$\|x\| = \sqrt{(-\frac{\sqrt{2}}{3})^2 + (\frac{5}{3})^2} = \sqrt{3}$.

Q15 $p = \langle x, u_2\rangle u_2 + \langle x, u_3\rangle u_3$

$= \frac{8}{3}u_2 - \frac{1}{\sqrt{2}}u_3$

$= (\frac{23}{18}, \frac{41}{18}, \frac{8}{9})$

$p - x = (\frac{5}{18}, \frac{5}{18}, -\frac{10}{9})$

$\langle p-x, u_2\rangle = \frac{2}{3}\times\frac{5}{18} + \frac{2}{3}\times\frac{5}{18} - \frac{1}{3}\times\frac{10}{9} = 0$

$\langle p-x, u_3\rangle = -\frac{1}{\sqrt{2}}\times\frac{5}{18} + \frac{1}{\sqrt{2}}\times\frac{5}{18} = 0$

Q16. $\langle u_1, x\rangle = c_1 = 4$

$\langle u_2, x\rangle = c_2 = 0$

$\|x\| = \sqrt{c_1^2+c_2^2+c_3^2} = 5$ $\therefore c_3 = \pm 3$

Q17 $v_1 = (\frac{1}{3}, \frac{2}{3}, -\frac{2}{3})^T$

$u_2 - \langle u_2, v_1\rangle v_1 = (4,3,2)^T - 2(\frac{1}{3}, \frac{2}{3}, -\frac{2}{3})^T = (\frac{10}{3}, \frac{5}{3}, \frac{10}{3})^T$

$v_2 = (\frac{2}{3}, \frac{1}{3}, \frac{2}{3})^T$

$u_3 - \langle u_3, v_1\rangle v_1 - \langle u_3, v_2\rangle v_2$

$= (1,2,1)^T - (\frac{1}{3}, \frac{2}{3}, -\frac{2}{3})^T - 2(\frac{2}{3}, \frac{1}{3}, \frac{2}{3})^T = (-\frac{2}{3}, \frac{2}{3}, \frac{1}{3})^T$

$v_3 = (-\frac{2}{3}, \frac{2}{3}, \frac{1}{3})^T$

Q18. (a) $v_1 = (\frac{2}{3}, \frac{1}{3}, \frac{2}{3})^T$

$u_2 - \langle u_2, v_1\rangle v_1 = (1,1,1)^T - \frac{5}{3}(\frac{2}{3}, \frac{1}{3}, \frac{2}{3})^T = (-\frac{1}{9}, \frac{4}{9}, -\frac{1}{9})^T$

$v_2 = (\frac{-1}{3\sqrt{2}}, \frac{2\sqrt{2}}{3}, \frac{-1}{3\sqrt{2}})^T$ v_1, v_2 are the orthonormal basis

(b) $\begin{pmatrix}2\\1\\2\end{pmatrix} = 3v_1$ $\begin{pmatrix}1\\1\\1\end{pmatrix} = \frac{5}{3}v_1 + \frac{\sqrt{2}}{3}v_2$

$\therefore A = \begin{pmatrix}\frac{2}{3} & -\frac{1}{3\sqrt{2}}\\ \frac{1}{3} & \frac{2\sqrt{2}}{3}\\ \frac{2}{3} & -\frac{1}{3\sqrt{2}}\end{pmatrix}\begin{pmatrix}3 & \frac{5}{3}\\ 0 & \frac{\sqrt{2}}{3}\end{pmatrix}$

(c) $A^TA = \begin{pmatrix}2&1&2\\1&1&1\end{pmatrix}\begin{pmatrix}2&1\\1&1\\2&1\end{pmatrix} = \begin{pmatrix}9&5\\5&3\end{pmatrix}$ $A^Tb = \begin{pmatrix}2&1&2\\1&1&1\end{pmatrix}\begin{pmatrix}12\\6\\16\end{pmatrix} = \begin{pmatrix}66\\36\end{pmatrix}$

$\begin{pmatrix}9&5\\5&3\end{pmatrix}x = \begin{pmatrix}66\\36\end{pmatrix}$ $x = \begin{pmatrix}9\\-3\end{pmatrix}$